

Java Programming Tasks & Instructions — CodeAlpha

Java Programming — Internship Overview

This internship program offers in-depth experience in Java development and object-oriented programming. CodeAlpha is a leading software development company focused on building robust, scalable, and enterprise-grade applications. The internship empowers students to master Java fundamentals, data structures, multithreading, and file handling, along with frameworks like Spring and Hibernate. Interns will work on real-world projects, gaining hands-on experience in backend development, API creation, and application deployment. With expert mentorship and a collaborative environment, interns will develop strong programming and problem-solving skills in Java.

Internship Perks

- Internship Offer Letter
 - Completion Certificate (QR Verified)
 - Unique ID Certificate
 - Letter of Recommendation (based on performance)
 - Job Opportunities / Placement Support
 - Resume Building Support
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Instructions for Interns

1. Share your internship status on **LinkedIn**, tagging **@CodeAlpha**.
 2. Complete the **assigned projects** within the mentioned time frame.
 3. Upload your complete source code to **GitHub** in a repository named:
`CodeAlpha_ProjectName`
 4. Post a **video explanation** of your project on LinkedIn with GitHub repo link.
 5. Submit your completed task using the **Submission Form**.
 6. **Complete any 3 or 2 out of the 4 tasks** listed below (from your domain).
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Java Programming Task List

(Complete any 2 or 3 of the following tasks)

TASK 1: Student Grade Tracker

- Build a Java program to **input and manage student grades**.
 - Calculate **average, highest, and lowest scores**.
 - Use **arrays or ArrayLists** to store and manage data.
 - Display a summary report of all students.
 - Make the interface **console-based or GUI-based** as desired.
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✓ TASK 2: Stock Trading Platform

- Simulate a basic **stock trading environment**.
 - Add features for **market data display** and **buy/sell operations**.
 - Allow users to track **portfolio performance** over time.
 - Use **Object-Oriented Programming (OOP)** to manage stocks, users and transactions.
 - Optionally, include **file I/O or database** to persist portfolio data.
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✓ TASK 3: Artificial Intelligence Chatbot

- Create a **Java-based chatbot** for interactive communication.
 - Use **Natural Language Processing (NLP)** techniques.
 - Implement **machine learning logic or rule-based answers**.
 - Train the bot to respond to frequently asked questions.
 - Integrate with a **GUI or web interface** for real-time interaction.
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✓ TASK 4: Hotel Reservation System

- Design a system to **search, book, and manage hotel rooms**.
 - Add **room categorization** (e.g., Standard, Deluxe, Suite)
 - Allow users to **make and cancel reservations**.
 - Implement **payment simulation and booking details view**.
 - Use **OOP + database/File I/O** for storing bookings and availability.
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Important Note

Internship Completion Criteria:

To be eligible for the internship certificate, participants must complete a minimum of **two or three tasks**. Submitting only one task will be considered **incomplete** and certificates will **not** be issued in such cases.



Submission Details

A submission form will be shared in your respective **WhatsApp group**. You are required to submit your completed task only through that form. Please follow the instructions mentioned in the form carefully to ensure your submission is accepted.



Contact Information

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